Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy under stated assumptions.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

With GMm=105 J·m, move an object from r=3 to r=7 m. Find work by gravity using U=-GMm/r.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mg Δh = ½mv²; v = √(2 g Δh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest 0.8 m above its lowest point reaches speed √(2 gh)=4 m/s there with g=10 and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Initial **Transfers** **Final**

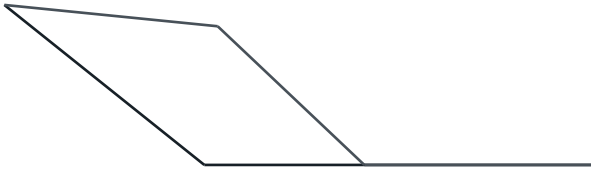
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two frictionless tracks start and end at the same heights but take different paths. Compare final speeds from rest.

Same start



Mark the vertical height change.

Same finish

Marzano check - circle: ① ② ③

04 Apply the physics

A coaster enters height 8 m at speed 6 m/s. Find speed at height 3 m with no losses. Use g=10. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

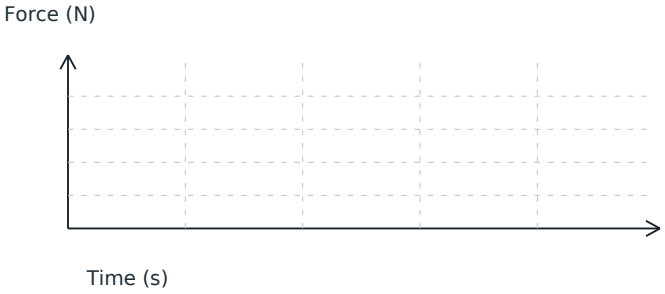
05 Explain your reasoning

A classmate uses total ramp length as h in mgh. Explain how to identify the correct measurement.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A positive triangular pulse peaks at 60 N and lasts 0.4 s. It acts on a 4 kg cart initially at -3 m/s. Find impulse and final velocity.
Show the signed impulse on the force-time model.



Marzano check - circle: 1 2 3

07 Exit check

A cart begins at rest at height 5 m. Can it reach height 6 m without an external energy input? Justify. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A frictionless coaster starts from rest at height 8 m above the bottom of a vertical loop of radius 2 m. Use g=10. Find speed and normal force per unit mass at the top; check contact. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3